

ARAVIND UNNI M S

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PROFESSIONAL SUMMARY

M.Tech Data Science candidate with 1.5 years of experience building production-grade AI systems, scalable Python backend services, and automated MLOps pipelines. Expertise spans PyTorch and TensorFlow, NLP (LLMs, RAG), AI agent and Agentic workflows. Proven track record of deploying robust AI solutions on AWS via Docker and FastAPI to reduce overhead and improve reliability. Seeking a Data Science or AI Engineering role to drive measurable business impact.

WORK EXPERIENCE

AI Research Intern – *CorrelationLab*
Kollam, India

October 2025 – April 2026

- Developed a predictive maintenance system using transfer learning on Short-Time Fourier Transform spectrograms, classifying 10 bearing fault states with 95 percent accuracy.
- Applied Grad-CAM explainability techniques to validate deep learning model reliability, achieving a 76.07 percent Faithfulness Index score and meeting production interpretability requirements.
- Delivered a real-time industrial monitoring dashboard integrating WebSocket-based data streaming, a FastAPI REST API backend, Supabase PostgreSQL logging, and automated SendGrid email alerts, cutting mean response time for fault notifications from hours to under 60 seconds.

Python Developer Trainee – *AadilTech Solutions Pvt Ltd*
Kollam, India

July 2023 – July 2024

- Architected a RESTful API using FastAPI and SQLAlchemy to fully digitize cargo lifecycle management, incorporating an auto-manifesting algorithm and programmatic PDF generation.
- Delivered an OTP-based driver verification portal for proof of delivery workflows, leveraging FastAPI Background Tasks for asynchronous email notifications..

PROJECTS

Production Grade voice AI agent for customer support

- Engineered a conversational AI pipeline using Llama 3.3-70B and LangChain, integrated with a Pinecone vector store and hybrid search for high-accuracy context retrieval.
- Connected Twilio for live call management, utilizing faster whisper for low latency Speech to Text (STT) and Edge TTS for responsive, natural voice synthesis.
- Implemented LangSmith for end-to-end tracing and continuous evaluation of the LLM workflows

Climate AI assistant using Databricks

- developed highly scalable RAG architecture on Databricks, processing complex climate datasets to ground Large Language Model responses
- Utilized the Databricks ecosystem (MLflow) to track experiments, manage prompt templates,
- Deployed the application on AWS (EC2, ECR) via Docker, delivering sub-second response latency to concurrent users through a FastAPI backend and a custom HTML, CSS, and JavaScript frontend.

Automated Stock Market Portfolio Optimization and Forecasting

- Constructed a real time asset price forecasting engine using Facebook Prophet, dynamically maximizing the Sharpe Ratio through Markowitz Modern Portfolio Theory, generating portfolio rebalancing recommendations with demonstrably improved risk-adjusted returns.
- Designed a production MLOps continuous integration and continuous deployment pipeline using CircleCI and GitHub Actions, automating daily data ingestion from yfinance, model inference, and synchronization to a PostgreSQL database on Supabase, saving time of daily manual intervention.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++.
- **Natural Language Processing and Generative AI:** BERT, NLTK, SpaCy, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, MCP, Agentic Workflows, Prompt Engineering, LLM Evaluation, LoRA, PEFT
- **Backend and API Development:** FastAPI, REST API, SQLAlchemy, WebSocket, Asynchronous Programming

- **MLOps and Cloud:** AWS (EC2, ECR), Databricks, Docker, MLflow, Git, GitHub, CI/CD (CircleCI, GitHub Actions), Power BI

EDUCATION

Master of Technology (M.Tech) in Data Science <i>Amrita School of Artificial Intelligence, Coimbatore</i> — CGPA: 8.15	2024 – 2026
Bachelor of Technology (B.Tech) in Civil Engineering <i>College of Engineering Trivandrum, Trivandrum</i>	2019 – 2023

CERTIFICATIONS AND ACHIEVEMENTS

- **Generative AI Mastery with Real-Time Projects** – Udemy (2024)
- **HackerRank Profile** - [Link](#)
- **Github Repository** - [Link](#)
- **Springer Nature Research Publication (2026)** – Co-authored peer-reviewed paper in Scientific Reports on AI-driven compressive strength prediction for geopolymer concrete [[View Publication](#)]